

Measurement & Probability

Collapse a superposition a thousand times and watch the histogram converge on $|\alpha|^2$.
Measurement is destructive, and that is the point.

EXAMPLE NOTES

Written by the QuantaVerse team to show what a module's notes can look like: each lesson's theory in brief, every formula typeset and numbered, and the key ones gathered on a formula sheet at the end. Professors can upload their own notes for this module on its page, and students will see them alongside these.

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CONVENTIONS

Multi-qubit states are written the way Qiskit prints them, highest qubit on the left: $|q_{n-1} \cdots q_1 q_0\rangle$, so the string 011 means $q_0 = 1, q_1 = 1, q_2 = 0$. N is a number of shots and p an outcome probability.

LESSON 2.1

The Born rule

KEY IDEA

A state is not observed directly. Measuring it returns outcome x with probability $|\langle x|\psi\rangle|^2$, and nothing more.

$$P(x) = |\langle x|\psi\rangle|^2 \quad (2.1.1)$$

the Born rule, for a measurement in the basis $\{|x\rangle\}$

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle \Rightarrow P(0) = |\alpha|^2, \quad P(1) = |\beta|^2 \quad (2.1.2)$$

for one qubit in the computational basis

The probabilities add to one precisely because the state is normalised. That is not a coincidence of notation: the normalisation condition of lesson 1.1 exists so that the Born rule yields a probability distribution.

$$\sum_x P(x) = \sum_x \langle \psi|x\rangle\langle x|\psi\rangle = \langle \psi|\psi\rangle = 1 \quad (2.1.3)$$

the outcomes are exhaustive

More generally a measurement is a set of *projectors* P_m , one per outcome, that are mutually orthogonal and sum to the identity. Measuring in a basis is the special case $P_x = |x\rangle\langle x|$.

$$p(m) = \langle \psi|P_m|\psi\rangle, \quad P_m P_{m'} = \delta_{mm'} P_m, \quad \sum_m P_m = I \quad (2.1.4)$$

projective measurement

Repeating a measurement many times gives averages. The average of an observable is its *expectation value*; for Z , whose outcomes are $+1$ for 0 and -1 for 1 , it is the difference of the two probabilities.

$$\langle Z \rangle = \langle \psi|Z|\psi\rangle = P(0) - P(1) \quad (2.1.5)$$

the expectation value of Z

WORKED EXAMPLE · READING A STATE

Let $|\psi\rangle = \frac{\sqrt{3}}{2}|0\rangle + \frac{1}{2}e^{i\pi/3}|1\rangle$.

$P(0) = \left|\frac{\sqrt{3}}{2}\right|^2 = \frac{3}{4}$ and $P(1) = \left|\frac{1}{2}e^{i\pi/3}\right|^2 = \frac{1}{4}$: the phase $e^{i\pi/3}$ has no effect on either.

$\langle Z \rangle = \frac{3}{4} - \frac{1}{4} = \frac{1}{2}$, which is also the height $r_z = \cos \theta$ of the state on the Bloch sphere, with $\theta = \pi/3$.

LESSON 2.2

Measuring changes the state

KEY IDEA

A measurement returns one outcome and leaves the state in the matching basis state. Measuring again only repeats the answer.

$$|\psi\rangle \mapsto \frac{P_m|\psi\rangle}{\sqrt{p(m)}} \quad (2.2.1)$$

the state just after outcome m is observed

For a computational-basis measurement of one qubit this says: whatever α and β were, the state afterwards is $|0\rangle$ or $|1\rangle$. The amplitudes are gone; all that remains is one bit.

$$|+\rangle \xrightarrow{\text{measure}} \begin{cases} |0\rangle & \text{with probability } \frac{1}{2} \\ |1\rangle & \text{with probability } \frac{1}{2} \end{cases} \quad (2.2.2)$$

a superposition, measured

A second measurement in the same basis repeats the first result with certainty, because a projector applied twice is the same projector. This is why a qubit cannot be read twice to learn more about the original amplitudes.

$$P_m^2 = P_m \Rightarrow p(m | m) = \frac{\langle\psi|P_m P_m P_m|\psi\rangle}{p(m)} = 1 \quad (2.2.3)$$

repeatability

Could the state be copied first, and each copy measured? No. The *no-cloning theorem* says no single operation copies every unknown state. The proof takes one line: an operation that copied two states would have to preserve their inner product, and squaring it at the same time.

$$U|\psi\rangle|0\rangle = |\psi\rangle|\psi\rangle \quad \forall \psi \Rightarrow \langle\phi|\psi\rangle = \langle\phi|\psi\rangle^2 \Rightarrow \langle\phi|\psi\rangle \in \{0,1\} \quad (2.2.4)$$

no-cloning: only orthogonal or identical states can be copied by one U

DEFINITION · DESTRUCTIVE MEASUREMENT

Measurement irreversibly turns amplitudes into one classical outcome. This is not a limitation to engineer away. Algorithms are built so that, just before the measurement, nearly all the amplitude sits on the answer.

WORKED EXAMPLE · WHAT IS LEFT

Measure $|\psi\rangle = \frac{1}{\sqrt{3}}|0\rangle + \sqrt{\frac{2}{3}}e^{i\pi/5}|1\rangle$ and get 1, which happens with probability $\frac{2}{3}$. The state is now exactly $|1\rangle$: the magnitude $\sqrt{2/3}$ and the phase $e^{i\pi/5}$ are both gone, and measuring again returns 1 with probability 1.

LESSON 2.3

Shot noise

KEY IDEA

Each shot is an independent draw. Counts follow a binomial distribution, whose spread shrinks only as $1/\sqrt{N}$.

Run the same circuit N times and count k , the number of times outcome 0 appears. Each shot is an independent trial with success probability p , so k is binomially distributed.

$$P(k) = \binom{N}{k} p^k (1-p)^{N-k}, \quad \mathbb{E}[k] = Np, \quad \sigma_k = \sqrt{Np(1-p)} \quad (2.3.1)$$

counts from N shots

$$\hat{p} = \frac{k}{N}, \quad \sigma_{\hat{p}} = \sqrt{\frac{p(1-p)}{N}} \quad (2.3.2)$$

the estimated probability, and its standard error

The error falls as $1/\sqrt{N}$. Halving it takes four times the shots; one more decimal place takes a hundred times.

SHOTS N	$\sigma_{\hat{p}}$ FOR $p = \frac{1}{2}$	TYPICAL READING
100	0.050	0.50 ± 0.05
1024	0.016	0.500 ± 0.016
10 000	0.005	0.500 ± 0.005
1 000 000	0.0005	0.5000 ± 0.0005

WORKED EXAMPLE · 1024 SHOTS OF A FAIR COIN

$$N = 1024, p = \frac{1}{2}: \sigma_k = \sqrt{1024 \cdot \frac{1}{2} \cdot \frac{1}{2}} = \sqrt{256} = 16.$$

So about 68% of runs land within 512 ± 16 and about 95% within 512 ± 32 . A result of 530 against 494 is entirely ordinary.

Getting *exactly* 512 is unlikely. For large N the central binomial probability is approximately $\sqrt{2/(\pi N)}$.

$$P\left(k = \frac{N}{2}\right) = \binom{N}{N/2} 2^{-N} \approx \sqrt{\frac{2}{\pi N}} \approx 0.025 \quad (N = 1024) \quad (2.3.3)$$

an exact split happens about one run in forty

Shot noise is not a hardware fault. A perfect, noiseless device still shows it, because it comes from sampling. Hardware noise adds *bias* on top, a shift that more shots do not remove.

LESSON 2.4

Changing the basis

KEY IDEA

Hardware measures in the Z basis only. To measure in any other basis, apply the inverse of the rotation that makes that basis, then measure as usual.

$$P(b_i) = |\langle b_i | \psi \rangle|^2 = |\langle i | U^\dagger | \psi \rangle|^2, \quad |b_i\rangle = U|i\rangle \quad (2.4.1)$$

measuring in the basis $\{U|i\rangle\}$ is applying $U^\dagger$, then measuring in $\{|i\rangle\}$

The X basis is made by H , and H is its own inverse, so an X measurement is a Hadamard followed by an ordinary measurement. The Y basis is made by SH , so its inverse is $HS^\dagger$: first $S^\dagger$, then H .

TO MEASURE IN	APPLY FIRST	OUTCOME 0 MEANS	OUTCOME 1 MEANS
Z basis	nothing	$ 0\rangle$	$ 1\rangle$
X basis	H	$ +\rangle$	$ -\rangle$
Y basis	$S^\dagger$, then H	$ +i\rangle$	$ -i\rangle$

$$HS^\dagger|+i\rangle = H \frac{|0\rangle + (-i)(i)|1\rangle}{\sqrt{2}} = H|+\rangle = |0\rangle \quad (2.4.2)$$

checking the Y-basis recipe

Bases like Z and X are *complementary*: a state that is certain in one is maximally uncertain in the other. The overlaps between their basis states all have the same size.

$$|\langle + | 0 \rangle|^2 = |\langle - | 0 \rangle|^2 = |\langle + | 1 \rangle|^2 = |\langle - | 1 \rangle|^2 = \frac{1}{2} \quad (2.4.3)$$

complementary bases: knowing Z tells you nothing about X

WORKED EXAMPLE · TELLING $|+\rangle$ FROM $|-\rangle$

Measured directly, both give 50/50, so no number of shots separates them.

Apply H first: $H|+\rangle = |0\rangle$ and $H|-\rangle = |1\rangle$. Now one shot is enough, with certainty.

This is how the Bloch vector of lesson 1.3 is estimated in practice: many shots measured directly give $\langle Z \rangle$, many after H give $\langle X \rangle$, and many after $S^\dagger H$ give $\langle Y \rangle$.

LESSON 2.5

Classical registers

KEY IDEA

A measurement writes a bit into a classical register. How those bits are ordered into a string is a convention, and Qiskit puts qubit 0 on the right.

$$|x\rangle = |q_{n-1} \cdots q_1 q_0\rangle, \quad x = \sum_{k=0}^{n-1} q_k 2^k \quad (2.5.1)$$

little-endian: q_0 is the lowest bit and is printed last

QUBITS	PRINTED STRING	INTEGER x
$q_0 = 1, q_1 = 1, q_2 = 0$	011	3
$q_0 = 0, q_1 = 0, q_2 = 1$	100	4
$q_0 = 1, q_1 = 0, q_2 = 1$	101	5

For several qubits the Born rule is unchanged: the probability of a whole string is the squared magnitude of its amplitude. The probability of *one* qubit's outcome is a marginal, found by summing over the other qubits.

$$P(q_k = b) = \sum_{x : x_k = b} |\alpha_x|^2 \quad (2.5.2)$$

the marginal probability of one qubit

WORKED EXAMPLE · A MARGINAL

$|\psi\rangle = \frac{1}{2}|00\rangle + \frac{1}{2}|01\rangle + \frac{1}{\sqrt{2}}|11\rangle$, written $|q_1 q_0\rangle$.

$P(q_0 = 1) = |\alpha_{01}|^2 + |\alpha_{11}|^2 = \frac{1}{4} + \frac{1}{2} = \frac{3}{4}$, and $P(q_1 = 1) = |\alpha_{11}|^2 = \frac{1}{2}$.

Circuit diagrams draw classical bits as double lines, because what they carry is ordinary information: it can be copied, stored and used to decide later gates.

DEFINITION · DEFERRED MEASUREMENT

A measurement whose result only controls later gates can be moved to the end of the circuit, with the classical control replaced by a quantum controlled gate. The output statistics are identical. The teleportation walkthrough relies on this.

$$\text{measure } q, \text{ then apply } U \text{ if the result is } 1 \equiv \text{controlled-}U \text{ from } q, \text{ then measure } q \quad (2.5.3)$$

the principle of deferred measurement

Keep the convention in mind when reading textbooks, which usually write the first qubit on the left: the same state is $|q_0 q_1\rangle$ there and $|q_1 q_0\rangle$ in Qiskit output.

MODULE 2 · FORMULA SHEET

Formula sheet

The key results of the module, numbered as they appear in the lessons.

the Born rule, for a measurement in the basis $\{|x\rangle\}$

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projective measurement

$$p(m) = \langle \psi | P_m | \psi \rangle, \quad P_m P_{m'} = \delta_{mm'} P_m, \quad \sum_m P_m = I \quad (2.1.4)$$

the expectation value of Z

$$\langle Z \rangle = \langle \psi | Z | \psi \rangle = P(0) - P(1) \quad (2.1.5)$$

the state just after outcome m is observed

$$|\psi\rangle \mapsto \frac{P_m |\psi\rangle}{\sqrt{p(m)}} \quad (2.2.1)$$

no-cloning: only orthogonal or identical states can be copied by one U

$$U|\psi\rangle|0\rangle = |\psi\rangle|\psi\rangle \quad \forall \psi \Rightarrow \langle \phi | \psi \rangle = \langle \phi | \psi \rangle^2 \Rightarrow \langle \phi | \psi \rangle \in \{0, 1\} \quad (2.2.4)$$

counts from N shots

$$P(k) = \binom{N}{k} p^k (1-p)^{N-k}, \quad \mathbb{E}[k] = Np, \quad \sigma_k = \sqrt{Np(1-p)} \quad (2.3.1)$$

the estimated probability, and its standard error

$$\hat{p} = \frac{k}{N}, \quad \sigma_{\hat{p}} = \sqrt{\frac{p(1-p)}{N}} \quad (2.3.2)$$

measuring in the basis $\{U|i\rangle\}$ is applying $U^\dagger$, then measuring in $\{|i\rangle\}$

$$P(b_i) = |\langle b_i | \psi \rangle|^2 = |\langle i | U^\dagger | \psi \rangle|^2, \quad |b_i\rangle = U|i\rangle \quad (2.4.1)$$

complementary bases: knowing Z tells you nothing about X

$$|\langle +|0\rangle|^2 = |\langle -|0\rangle|^2 = |\langle +|1\rangle|^2 = |\langle -|1\rangle|^2 = \frac{1}{2} \quad (2.4.3)$$

little-endian: q_0 is the lowest bit and is printed last

$$|x\rangle = |q_{n-1} \cdots q_1 q_0\rangle, \quad x = \sum_{k=0}^{n-1} q_k 2^k \quad (2.5.1)$$

the marginal probability of one qubit

$$P(q_k = b) = \sum_{x: x_k = b} |\alpha_x|^2 \quad (2.5.2)$$